TBO VOYAGEHACK 4.0 · MULTI SUPPLIER OPTIMIZATION · TEAM ALPHAFIRE

SupplyLens

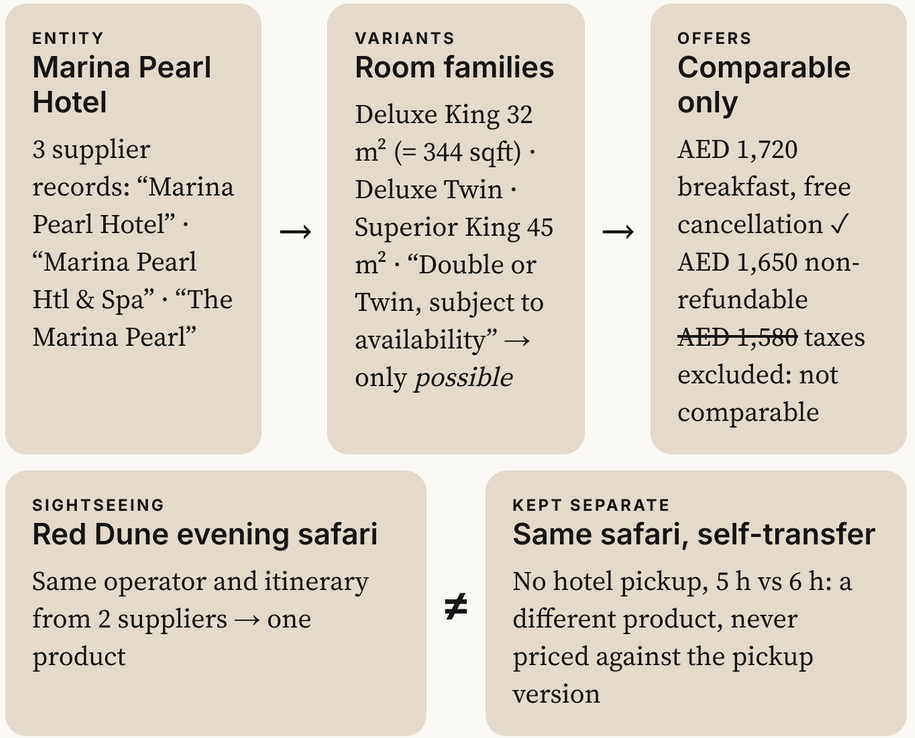
Make hotel and sightseeing inventory comparable, without losing what makes each option different.

CONTENT 1 OF 2 · POTENTIAL IMPACT AND INNOVATION

One trustworthy listing per product. Every meaningful option kept.

THE PROBLEM

The same hotel arrives from several suppliers under different names, so listing pages fill with duplicates. Naive deduplication goes the other way: it merges a pickup tour with a self-transfer one, or compares a tax-exclusive price with a full one.



IMPACT

- Cleaner listing pages: one card per product, with supplier and variant counts
- Price competitiveness: the best *eligible* offer, not the lowest headline
- Fewer wrong merges, so fewer post-booking surprises

INNOVATION

- Variant-safe consolidation
- Traceable reasons and a reviewer path
- One model for both verticals

Synthetic data. All suppliers, prices and availability are fictional.

CONTENT 2 OF 2 · IMPLEMENTATION, TECHNICAL FEASIBILITY AND ADOPTION RISK

Simple, deterministic, explainable, and safe to adopt.

1 · Normalize
names, units, phones, addresses, FX, with source kept

2 · Match
match / review / distinct with hard conflicts

3 · Map variants
rooms by occupancy, bed, size, view

4 · Compare offers
eligibility first: tax, freshness, policy

5 · Rank and explain
intent, value, policy fit, reliability

6 · Review
uncertain pairs, reversible, audited

FEASIBILITY

- Python + FastAPI, no heavy infrastructure, millisecond searches
- Rules and similarity over fields suppliers already send; no model training needed
- An early end-to-end prototype already runs on synthetic data for both verticals

PROOF PLAN (PER VERTICAL)

Auto-link precision, harmful-merge suite, offer-correctness cases, and NDCG against raw-feed and cheapest-price baselines. Early synthetic check: **0** harmful merges vs **3** for name-only matching (9 labelled pairs each).

ADOPTION RISKS AND ANSWERS

False merges	Hard conflicts block them; mixed evidence goes to a review queue; every decision is reversible
Missing data	Unknown stays unknown, with field-level sources; never shown as “included”
Stale or partial prices	Excluded from “best”, with the reason shown
Integration	Offline evaluation on TBO sample feeds, then a controlled pilot

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THANK YOU

Comparable, not flattened.

Team AlphaFire · SupplyLens

`github.com/gurlivsingh/supplylens`

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